

Project Documentation:

AI-Powered Moodboard Generator (MVP)

Group : **D_For_Design**

Team Members

Daphal Sanket Anil - 210300

Gaurav Kumar - 220404

Sachin Kumar Meena - 210889

Rishikesh Meena - 210855

Raman kumar - 210824

November 2025

Abstract

This document provides a detailed overview of the Minimum Viable Product (MVP) for the **AI-Powered Moodboard Generator**. The project is designed to automate the initial visual ideation phase of a design process by converting natural language briefs into actionable, structured design elements. The core functionality includes Natural Language Processing (NLP) for keyword extraction, rule-based algorithms for generating aesthetically coherent color palettes, and the creation of a definitive **Visual Strategy Blueprint** — a refined, copy-paste prompt containing explicit color codes and style rules ready for a premium AI image generation service (like DALL-E or Imagen). This MVP successfully demonstrates the automated translation of creative intent into technical design specifications.

Contents

Section	Topic	Page
1	Strategic Rationale & Approach	1
1.1	Problem: Bridging Intent and Execution	1
1.2	Solution: The AI-Driven Blueprint	1
2	Project Features & Implementation	2
2.1	Natural Language Processing (NLP) Core	2
2.2	Rule-Based Style and Color Generation	2
2.3	Visual Strategy Blueprint Output	3
3	Technology Stack	3
4	Project Impact & Conclusion	4
5	Appendix: Project Links	4

1 Strategic Rationale & Approach

1.1 Problem: Bridging Intent and Execution

In creative workflows, defining a visual moodboard is often iterative and subjective, relying on designers manually searching for inspiration and defining core aesthetics. The primary challenges identified were:

- **Subjectivity in Color Choice:** Manually selecting a palette that is both relevant to the brief and aesthetically harmonious.
- **Inefficient Keyword Generation:** Translating a long narrative brief into concise, powerful keywords required by modern AI image generators.
- **Tool Fragmentation:** Needing separate tools for color, texture, and image generation.

The project aims to solve this by providing a single tool that automates the core translation layer between human creative language and AI-readable design specification.

1.2 Solution: The AI-Driven Blueprint

The **AI-Powered Moodboard Generator** functions as a translation engine. Instead of generating final assets, the MVP focuses on producing a **Visual Strategy Blueprint** — the high-value output containing all necessary parameters: color codes, style classification, and a hyper-refined prompt for any third-party AI image generation service.

2 Project Features & Implementation

2.1 Natural Language Processing (NLP) Core

The foundation of the application is its ability to parse and understand creative intent from unstructured text.

- **Mechanism (analyze_brief):** Uses the `spaCy` library with `en_core_web_sm` for tokenization and POS tagging.
- **Extraction:** Extracts key **Nouns** and **Adjectives**, filtering out stop words using `nltk.corpus`.
- **Classification:** Keywords are matched against a predefined `STYLE_MAPPING` dictionary to determine the dominant visual category.

2.2 Rule-Based Style and Color Generation

Once classified, deterministic algorithms generate consistent and aesthetic color palettes.

- **Style Mapping (STYLE_MAPPING):** Defines 9 aesthetic categories with hue ranges, saturation, and visual keywords.
- **Palette Generation (generate_palette_from_hsv):** Uses the `colorsys` library to create a 5-color palette harmonizing with the desired mood.

2.3 Visual Strategy Blueprint Output

This section serves as the final, actionable deliverable for downstream AI tools.

- **Structured Prompt Generation:** Combines user input, mapped style keywords, and generated hex color codes.
- **Explicit Color Inclusion:** Lists 5 generated color codes (e.g., #6d9bf2, #6567cb, ...) for visual coherence.
- **Clean Presentation:** Final output shown in a clear, copy-pasteable format.

Example AI Prompt Output:

```
Generate an abstract design concept for a Modern / Geometric moodboard.  
Key stylistic elements: clean, line, glass, steel, abstract.  
Original theme: book cover of crime thriller novel.  
Primary colors to include are the hex codes:  
#6d9bf2, #6567cb, #725ba5, #6d4c7f, #583a59.  
Focus on texture, composition, and lighting.
```

3 Technology Stack

Area	Technology	Rationale
Frontend	Streamlit	For rapid development of interactive web applications.
Backend	Python 3.9+	Core engine for NLP and algorithmic processing.
NLP Engine	spaCy (<code>en_core_web_sm</code>)	Efficient keyword and POS extraction.
Dependencies	NLTK, NumPy, colorsys	For text processing and HSV-to-RGB conversion.

4 Project Impact & Conclusion

The MVP validates automated creative translation. It reduces manual effort, ensures consistency between color palette and intended mood, and integrates seamlessly with AI image generators — cutting down ideation time and accelerating concept development.

5 Appendix: Project Links

- **GitHub Repository:** [github](#)